

Microbiome Lab Report

Purpose of Assignment

Being a scientist means a) applying the scientific method to understand a system, b) analyzing data from experiments, and c) sharing your conclusions. A lab report is the traditional way that scientists communicate their findings. Lab reports follow a defined format and style to make them more broadly useful.

With this assignment, you have a unique opportunity to contribute to an active research study at the University of Michigan called the Michigan Microbiome Project (IRB: HUM00166150). Human gut diseases, such as colon cancer and inflammatory bowel disease, are not well understood. You are contributing to an investigation into the function of the human gut microbiome to help scientists develop new probiotics and therapies that will improve human health. More specifically, scientists are interested in learning how our diet can affect the gut microbiome. In BIO 173, our class will (1) identify gut bacteria from prior participant fecal samples using molecular techniques and DNA sequencing, and (2) investigate how timing of breath samples reflects activities of the gut microbiome. All of the data we collect and analyze will be submitted to the researchers of the Michigan Microbiome Project.

As you complete the Microbiome experiments, this Lab Report will allow you to showcase the following scientific skills:

- ❖ **Scientific Method:** Explain how this experiment is set up to test a specific hypothesis through appropriate methods
- ❖ **Disciplinary Knowledge:** Describe the function of gut microbes; describe how exhaled gases can be used to measure gut fermentation; identify gut bacteria using DNA sequencing
- ❖ **Data Analysis:** Analyze and interpret data from your experiment and highlight results that respond to your objective and hypothesis
- ❖ **Scientific Communication:** Apply the model & conventions of a standard Lab Report format to communicate your hypothesis, experiment, and conclusions to a scientific audience.

The structure of a Lab Report is one form for efficiently transmitting results to a scientific audience. It is also a form of persuasive essay in that you are laying out a question, presenting data that address the topic and then making the case for a certain answer to the question. When you write your lab report, please follow the format and instructions outlined below.

Report Format

- No more than 4 pages (not including tables and figures, or resources cited)
- Covers just the breath gas experiment (sequence ID submitted separately)
- Double spaced with 1-inch margins
- Organized and labeled by sections: Title, Introduction, Methods, Results, Discussion, Resources Cited
- Submit your lab report in PDF or Microsoft Word format

Tasks & Assignment Components

A standard lab report has 4 main sections that work together within a defined structure: **Introduction**, **Methods**, **Results**, and **Discussion**. Other key components are the **Title** and a final **Resources Cited** section. As you draft each section, think first about the *Purpose* of that section in the context of your Lab Report and then attend to the *Expectations* that outline what to include. Your complete Lab Report should be a maximum of 4 pages (double-spaced with 1-inch margins).

TITLE

Purpose: The Title should succinctly convey the topic and main result of the study to let potential readers decide whether to dig into the full version.

ABSTRACT

Purpose: Offer a longer (one paragraph) summary of the paper for the reader's convenience.

Note: *You will not have to write an abstract for this assignment. For one thing, the paper is short already. For another, you had practice writing an abstract with the molecular biology paper.*

INTRODUCTION (~1 page) - Context for significance of experiment & findings

Purpose: The introduction provides background information and the state of existing knowledge on a subject, leading to an unanswered question that is the focus of the current work. There should be a clear chain of connection proceeding from the general to the specific. The Introduction should clearly state the objective of the study and the hypothesis being tested. A good Introduction will provide necessary context for the conclusions that will be presented later in the Report.

Expectations

BACKGROUND INFORMATION

- Drawing on lecture and lab notes, explain what is the gut microbiome, its importance and its impacts on human health
- Using lecture and lab notes, explain how diet influences the gut microbiome and how the microbiome contributes to our breath gases.
 - how do gut microbes process the foods we eat (i.e., fermentation)?
 - how do products of microbial metabolism end up in our exhaled breath? gases are released?

OBJECTIVES & HYPOTHESES

- Clearly state the objective of the study that was described in lecture
- Clearly state the hypothesis that was provided in lecture

METHODS (~0.5 - 1 page) - Experimental setup

Purpose: The Methods section should give a brief overview of the procedures used in the experiment. Clear procedures and methods are important for providing readers with the context to assess and interpret the results. While this section should provide enough detail so that another scientist could repeat the experiment, it is NOT designed to cover step-by-step methods on commonly understood techniques. The methods should answer the questions how, where and when. (e.g., the conditions for PCR, not the protocol from the lab manual). Where a particular mode of analysis will be used (e.g., statistical parameters) or where error correction is applied, this should be explained in the results section.

Expectations

OVERVIEW OF PROCEDURE AND EXPERIMENTAL SETUP

- Detail the timing of samples in this study.
- Clearly describe how fasting breath samples were measured (i.e., Quintron breath analyzer)

STATISTICAL ANALYSIS

- Clearly and concisely describe the data analysis you performed, including mean, variance, t-tests, or any other statistical analyses performed in your study
- Describe the criteria for determining data were in error

Note: DO NOT COPY the lab manual. State the methods in YOUR OWN WORDS.

RESULTS (~0.5 - 1 page, not including figures) – Report the numbers

Purpose: The Results section presents what was found without requiring the reader to interpret data from figures and/or tables. It also organizes the results into a narrative for the reader and highlights the key takeaways. You can think of your results as ‘just the facts’. The key to making the results section impactful to the reader is often condensing many numbers into a concise table or graph.

Expectations

SUMMARIZE the data and observations that are important to address your hypotheses, including the following:

- Use the class breath data to describe the relationship between sampling time and breath gases.
 - What happens to the level of hydrogen gas in the breath after a meal? Was it significant
 - What happens to the level of methane gas in the breath after a meal? Was it significant?
 - Are there more people with a methane-positive result after a meal?
- Provide the key statistics in a table (i.e., means and p-values).
- Can you say anything about the time needed to see fiber effects?
- Are there any other observations you can draw out of the data?

PROPER FIGURES AND TABLES follow these rules

- Provide tables or figures that summarize the data in a clear and informative way. Do NOT copy tables directly from the raw data file.
- Tables and figures are clearly labeled (e.g., Table 1 or Figure 1) and include a descriptive caption. Table captions are always *above* a table, while figure captions are placed *below* a graph, picture, or sketch.
- Axes in tables and figures are properly labeled, including units of measurement.

- Tables and figures should be placed at the end of your lab report and appropriately referenced in the text. Example: *The amount of hydrogen in fasting breath samples increased when fiber was increased in the diet (Figure 1).*
- See below for examples of a correctly formatted table and figure

Note: Do not include ALL results, only include those results that you can directly use to support or refute the hypothesis. The Results section should NOT include interpretation - save this for the Discussion.

DISCUSSION (~1-1.5 pages) - Interpretation & Conclusions

Purpose: This is the section where you explain what your results *mean* and interpret the data in relation to your original hypothesis. It should be the longest section and the one for which you spend the most time and effort. The 'closing statement' is where you make your case.

Expectations

STATE CONCLUSIONS & ADDRESS HYPOTHESIS

- Expand on the results observed. Offer explanations and context. State whether they support or refute the hypothesis in the Introduction.
 - Does timing of sampling affect gases in the breath? Why or why not?
 - What are differences between the two gases measured and how might that influence the results?

INFLUENCES ON RESULTS [Reflection] - Please reflect on the following questions in your discussion:

- What other factors might affect breath gases, and why?
- Are those factors affecting the results of this experiment or do they all average out?

APPLICATION TO THE NATURAL WORLD [Reflection] - Please reflect on the following questions:

- Describe how the relationship between fiber in the diet and gases in the breath relates to the health of the gut microbiome
- Explain how the gases in the breath might be used as a diagnostic tool for monitoring human health. What recommendations would you make about timing of samples?
- Besides fiber, what other nutrients might be useful to examine when investigating the function and composition of the gut microbiome?
- What next research steps would you recommend scientists take for this research project?

SOURCES OF ERROR & FUTURE DIRECTIONS

- Speculate on potential sources of error and limitations of the experimental design
- How can the experiment be improved/designed differently

Note: Make sure this is genuinely interpretive, not just a restatement of the Introduction or Results sections.

RESOURCES CITED - Supporting References

Purpose: Serves to provide readers with any additional results that support (or argue against) your conclusions. This section can also be thought of as a kind of database, where readers can find key resources that provide context for the topic of your study. References are most often cited to broaden the Introduction and Conclusion or explain Methods developed by others. And finally, this section is important because it gives professional credit to scholars whose work you are relying and building upon.

Expectations

- You must cite the lab and lecture material in your lab report. Please cite these materials in the same format as the following examples:
 - **Lab Manual in-text citation:** The 16s rRNA was amplified using PCR and prepared for DNA sequencing (BIO 173 Microbiome Lab Manual, 2024).
 - **Lab Manual in Resources Cited:** *BIO 173 Microbiome Lab Manual*. (2024). University of Michigan, Ann Arbor, MI
 - **Lecture in-text citation:** The 16s rRNA was amplified using PCR and prepared for DNA sequencing (Ammerlaan 2024).
 - **Lecture in Resources Cited:** Ammerlaan, M., (2024). *Lecture on Microbiome 1*. University of Michigan, Ann Arbor, MI
- You are also required to find and incorporate 1 primary research article outside the sources provided in class. You may use this to broaden your Introduction or bolster your Conclusions. You should cite the reference in text just as you would for the Lab Manual. The full reference should be provided in the Resources Cited section at the end of the paper. Examples:

In-text citation: Invasive mussels experience an increased catalase response when exposed to extreme temperatures (Nowicki & Kashian, 2018).

In Resources Cited: Nowicki CJ and DR Kashian. 2018. Comparison of lipid peroxidation and catalase response to single and multiple stressors in invasive dreissenid mussels. *Environmental Toxicology and Chemistry*, 37(6), 1643-1654

Examples of figures (caption and formatting):

Table 1. Number of seed types captured by different beak types as part of natural selection and evolution laboratory exercise, University of Michigan, Fall 2021.

Seed type	Beak type					χ^2 test	
	Spoon	Knife	Fork	Forceps	Spatula	χ^2	P
Sunflower	18	10	15	19	10	11.50	<0.05
Split pea	8	4	10	12	8	35.40	<0.001
Grass seed	5	0	2	5	1	76.75	<0.001
Pinto beans	24	15	29	50	20	51.10	<0.001
Safflower	11	7	8	17	7	28.60	<0.001

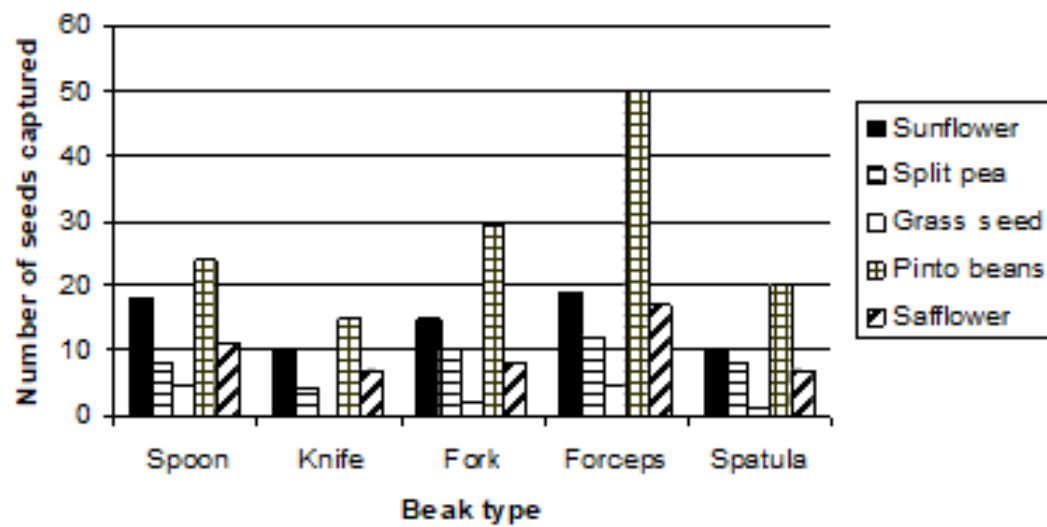


Figure 1. Number of seed types captured by different beak types as part of natural selection and evolution laboratory exercise, University of Michigan, Fall 2021.

Grading Rubric

Purpose: The following grading rubric synthesizes the expectations above into a roadmap for how we will be assessing your Lab Report. If you have questions, please refer to the details in the instructions above or speak with your GSI.

Category	Grading Scale - 20 Points Total		
TITLE 1 point possible	1 point Title declares the subject and major finding	0.5 point Title is vague or non-descriptive	0 points Title missing or not related to work
INTRODUCTION 3 Points Possible	2 points Introduction provides clear explanations of the necessary background on the gut microbiome, the link between the gut microbiome, diet, breath gases and the impacts of the microbiome on human health	1 point Introduction is vague or missing some of the necessary background information, definitions, or descriptions.	0 point Introduction is missing significant key information and descriptions.
	1 point Objective and hypothesis are clearly stated regarding the relationship between timing of breath samples and metabolic activity in the gut microbiome	0.5 points Objective and hypothesis are included but are unclear or not connected to the background given in the Introduction	0 points Either the objective or hypothesis is missing or is not appropriate for this experiment
METHODS 3 Points Possible	2 points Overview of procedure and experimental setup provides sufficient detail of methods and procedures in your own words	1 point Overview of procedures and experimental setup are provided but are missing important details	0 points There is minimal attempt at describing the experimental setup OR methods are copied from the lab manual
	1 point Description of data analysis is clear and concise and includes appropriate statistical tests and criteria for error handling	0.5 points Description of data analysis is missing a description of statistical tests or missing other major details	0 points There is no description of data analysis

RESULTS 4 Points Possible	2 points Summarizes the results and statistics that can be derived from the of class timing/breath analyses	1 point Analysis of the results and statistics is incomplete, not appropriately tied to the data, or partially wrong	0 points There is minimal description of the results; statistics are missing from the results section.
	2 points Tables & figures are clear, informative, properly labeled, and appropriately referenced in the text.	1 point Tables & figures are clear and informative but may not be properly labeled or referenced in the text.	0 points Tables and figures are missing or are directly copied from the raw data file.
DISCUSSION 4 Points Possible	1 point States how/why the results of the experiment justify the conclusions and explain whether they support or reject the hypotheses.	0.5 points Hypotheses are clearly supported/rejected but inadequate justification is provided	0 points Conclusions are stated but are not connected to the results. Hypotheses are supported/rejected with no justification.
	2 points Connects the results to our understanding of the gut microbiome including details about metabolism, other influences on results, and potential applications	1.5 points Partial effort to make larger connections.	1 point or less There may be minimal or no description of influences on the results, applications to the natural world and sources of error/limitations
	1 point Good plan for future work and identifying/correcting sources of error	0.5 points Plans and sources of error/limitations are explained but briefly	0 points Future plans are not well thought out or missing
LITERATURE 1 Point Possible	1 point Supporting resources are properly referenced in citations and end of report	0.5 points Improper format or place	0 points References missing
FORMAT 2 Points Possible	2 points Follows the standard structure and sections, within allotted page limit.	1 point Missing one of the formatting criteria; over the page limit by a half page	0 points Over the page limit by a page or more; or missing two or more of the formatting criteria
COHESION 2 Points Possible	2 points The report flows as a cohesive narrative and there are minimal spelling or grammatical errors	1 point There is some disconnect between sections of the report and there are some spelling or grammatical errors	0 point The report is missing an overall narrative arc that connects sections. There are numerous spelling or grammatical errors.